10.1.1 Energy band diagram of solids

In an isolated atom, the electronic energy levels are widely separated and are far apart and the energy of the electron is decided by the orbit in which it revolves around the nucleus. However, in the case of a solid, the atoms are closely spaced and hence the electrons in the outermost energy levels of nearby atoms influence each other. This changes the nature of the electron motion in a solid from that in an isolated atom to a large extent.

The valence electrons in an atom are responsible for the bonding nature. Let us consider an atom with one electron in the outermost orbit. It means that the number of valence electrons is one. When two such atoms are brought close to each other, the valence orbitals are split up into two. Similarly the unoccupied orbitals of each atom will also split up into two. The electrons have the choice of choosing any one of the orbitals as the energy of both the orbitals is the same. When the third atom of the same element is brought to this system, the valence orbitals of all the three atoms are split into three.The unoccupied orbitals also will split into three.

In reality, a solid is made up of millions of atoms. When millions of atoms are brought close to each other, the valence orbitals and the unoccupied orbitals are split according to the number of atoms. In this case, the energy levels will be closely spaced and will be difficult to differentiate the orbitals of one atom from the other and they look like a band as shown in Figure 10.2. **This band of very large number of closely spaced energy levels in a very small energy range is known as energy band.**

The energy band formed due to the valence orbitals is called **valence band (VB)** and that formed due to the unoccupied orbitals to which electrons can jump when energised is called the **conduction band (CB).** The energy gap between the valence band and the conduction band is called **forbidden energy gap (E_g).** Electrons cannot exist in the forbidden energy gap.

A simple pictorial representation of the valence band and conduction band is shown in Figure 10.2(a). E_V represents the maximum energy of the valence band and E_C represents minimum energy of the conduction band. The forbidden energy gap, $E_g = E_C - E_v$. We know that the Coulomb force of attraction between the orbiting electron and the nucleus is inversely proportional to the distance between them. Therefore, the electrons in the orbitals closer to the nucleus are strongly bound to it. Hence, the electrons closer to nucleus require a lot of energy to be excited. The electrons in the valence band are loosely bound to the nucleus and can be easily excited to become free electrons.

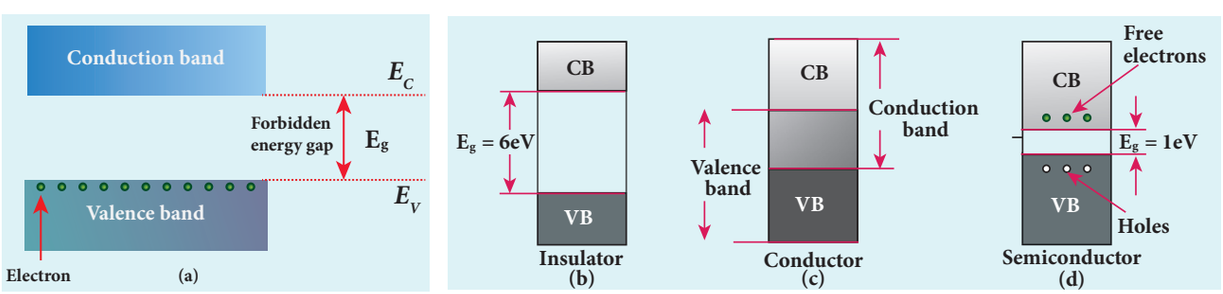


Figure 10.2 (a) Schematic representation of valence band, conduction band and forbidden energy gap. Energy band structure of (b) Insulator (c) Conductor (d) Semiconductor

XII_Physics_Vol-2_UNIT-10_Part 1_EM.indd 197 19-12-2022 16:38:47